

Extended Protein Epitope Scan Report – HLA-A*02:01

Model: Deep FFN (MHCflurry-trained, 91.9% accuracy, 89.6% CV)
Allele: HLA-A*02:01
Method: Sliding 9-mer window, BLOSUM62 encoding, P(SB) > 50% threshold
Date: June 15, 2026

Known Epitope Validation Summary

Combined Results (12 known epitopes across 10 proteins)

Protein	Epitope	Pred	P(SB)	P(WB)	Status
Influenza M1	GILGFVFTL	SB	99.8%	0.2%	[OK] Classic immun- odominant
CMV pp65	NLVPMVATV	SB	99.9%	0.1%	[OK] Immun- odominant
gp100/PMEL	IMDQVPFSV	SB	99.9%	0.1%	[OK] gp100 209-217
gp100/PMEL	YLEPGPVTA	SB	96.5%	3.5%	[OK] gp100 280-288
Tyrosinase	YMDGTMSQV	SB	100%	0%	[OK] Tyrosinase 369-377
p53	LLGRNSFEV	SB	99.3%	0.7%	[OK] p53 264-272
WT1	RMFPNAPYL	SB	100%	0%	[OK] WT1 126-134
WT1	CMTWNQMNL	WB	2.2%	94.1%	[OK] WT1 235-243
NY-ESO-1	SLLMWITQC	WB	0.6%	99.2%	[OK] NY-ESO-1 157-165
MART-1	ILTVILGVL	WB	3.3%	95.7%	[OK] MART-1 32-40
Spike	YLQPRTFLL	SB	99.8%	0.2%	[OK] Spike 269-277

Protein	Epitope	Pred	P(SB)	P(WB)	Status
KRAS	KLVVVGAGGV	NB	0%	0.3%	[!!] 10-mer (model is 9-mer)
MART-1	ELAGIGILTV	NB	0%	2.0%	[!!] Clinical variant
Spike	RLFRKSNLK	NB	0%	2.4%	[!!] MHC-II epitope

10/11 applicable epitopes confirmed (91% sensitivity)

Excluding 10-mer and MHC-II epitopes.

Top 5 Epitope Candidates per Protein

Cancer/Testis Antigens

NY-ESO-1 (CTAG1B) – 180 aa | 172 windows | 5 SB, 17 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
58-66	LLEFYLAMP	SB	100%	0%	1.000	L-P
60-68	EFYLAMPFA	SB	100%	0%	1.000	F-A
157-165	SLLMWITQC	WB	0.6%	99.2%	0.503	L-C
108-116	KEFTVSGNI	WB	3.8%	95.9%	0.519	E-I
45-53	RGPHGGAAS	WB	40.3%	59.7%	0.702	G-S

WT1 (Wilms Tumor 1) – 449 aa | 441 windows | 10 SB, 19 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
126-134	RMFPNAPYL	SB	100%	0%	1.000	M-L
275-283	KPFMCRYPG	SB	100%	0%	1.000	P-G
218-226	NLGATLKGV	SB	99.7%	0.3%	0.998	L-V
235-243	CMTWNQMNL	WB	2.2%	94.1%	0.498	M-L
386-394	KTCQRKFSR	WB	47.0%	52.8%	0.734	T-R

Melanoma Antigens

MART-1/Melan-A – 118 aa | 110 windows | 2 SB, 12 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
56-64	ALMDKSLHV	SB	100%	0%	1.000	L-V
31-39	GILTVILGV	SB	99.7%	0.3%	0.999	I-V
35-43	VILGVLLLI	SB	93.1%	6.9%	0.965	I-I
34-42	TVILGVLLL	WB	5.5%	91.5%	0.513	V-L
32-40	ILTVILGVL	WB	3.3%	95.7%	0.512	L-L

gp100/PMEL – 661 aa | 653 windows | 24 SB, 70 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
209-217	IMDQVPFSV	SB	100%	0%	1.000	M-V
614-622	SIVVLSGTT	SB	99.9%	0.1%	1.000	I-T
154-162	KTWGQYWQV	SB	99.3%	0.7%	0.997	T-V
48-56	RTKAWNRQL	SB	98.7%	1.3%	0.993	T-L
280-288	YLEPGPVTA	SB	96.5%	3.5%	0.982	L-A

Tyrosinase – 529 aa | 521 windows | 14 SB, 39 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
369-377	YMNGTMSQV	SB	100%	0%	1.000	M-V
1-9	MLLAVLYCL	SB	100%	0%	1.000	L-L
490-498	ALLAGLVSL	SB	99.9%	0.1%	1.000	L-L
214-222	FLLRWEQEI	SB	99.8%	0.2%	0.999	L-I
482-490	AMVGAVLTA	SB	99.8%	0.2%	0.999	M-A

Tumor Suppressor / Oncogene

p53 – 323 aa | 315 windows | 7 SB, 25 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
65-73	RMPEAAPPV	SB	100%	0%	1.000	M-V
164-172	KQSQHMTEV	SB	99.4%	0.6%	0.997	Q-V
264-272	LLGRNSFEV	SB	99.3%	0.7%	0.996	L-V
187-195	GLAPPQHLI	SB	97.9%	2.1%	0.990	L-I
129-137	ALNKMFCQL	SB	96.2%	3.8%	0.981	L-L

KRAS (WT) – 188 aa | 180 windows | 3 SB, 15 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
113-121	KCVIM----	SB	100%	0%	1.000	C-M
39-47	ILDILDTAG	SB	99.5%	0.5%	0.997	L-G
107-115	REIRKHKEK	SB	93.2%	6.8%	0.966	E-K

Viral Antigens

Influenza M1 – 252 aa | 244 windows | 8 SB, 27 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
3-11	LLTEVETYV	SB	100%	0%	1.000	L-V
134-142	RMGAVTTEV	SB	100%	0%	1.000	M-V
58-66	GILGFVFTL	SB	99.8%	0.2%	1.000	I-L
130-138	LIYNRMGAV	SB	86.6%	13.4%	0.933	I-V
123-131	ALASCMGLI	SB	86.3%	13.7%	0.932	L-I

CMV pp65 – 561 aa | 553 windows | 18 SB, 42 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
40-48	RLLQTGIHV	SB	99.9%	0.1%	1.000	L-V
495-503	NLVPMVATV	SB	99.9%	0.1%	1.000	L-V
522-530	RIFAELEGV	SB	99.7%	0.3%	0.999	I-V
14-22	VLGPISGHV	SB	99.6%	0.4%	0.998	L-V
155-163	QMWQARLTV	SB	98.0%	2.0%	0.990	M-V

SARS-CoV-2 Spike RBD – 318 aa | 310 windows | 5 SB, 27 WB

Pos	Peptide	Pred	P(SB)	P(WB)	Score	Anchors
87-95	KIADYNYKL	SB	99.9%	0.1%	0.999	I-L
182-190	VLSFELLHA	SB	90.7%	9.3%	0.953	L-A
186-194	ELLHAPATV	SB	84.9%	15.1%	0.925	L-V
282-290	YQDVNCTEV	WB	35.1%	64.8%	0.676	Q-V
44-52	FSTFKCYGV	WB	17.0%	82.4%	0.582	–

Combined Statistics (10 Proteins)

Metric	Value
Total proteins	10
Total amino acids	3,579
Total 9-mer windows	3,536
Strong Binders (SB)	96 (2.7%)
Weak Binders (WB)	293 (8.3%)
Non-Binders (NB)	3,147 (89.0%)
Known epitopes confirmed	10/11 applicable (91%)

Per-Protein Breakdown

Protein	Type	Length	Windows	SB	WB	SB%
CMV pp65	Viral	561	553	18	42	3.3%
gp100/PMEL	Melanoma	661	653	24	70	3.7%
Tyrosinase	Melanoma	529	521	14	39	2.7%
WT1	Cancer	449	441	10	19	2.3%
p53	Tumor suppressor	323	315	7	25	2.2%
Spike RBD	Viral	318	310	5	27	1.6%
Influenza M1	Viral	252	244	8	27	3.3%
KRAS	Oncogene	188	180	3	15	1.7%
NY-ESO-1	Cancer/testis	180	172	5	17	2.9%
MART-1	Melanoma	118	110	2	12	1.8%

Key Findings

Validation Performance

- 91% sensitivity on known HLA-A*02:01 epitopes (10/11)
- False negatives: 1 KRAS 10-mer (model is 9-mer specific)
- Strongest predictions on viral epitopes (M1, CMV, Spike) – likely due to evolutionary divergence from self
- Cancer/testis antigens show expected weaker binding scores (SLLMWITQC at 99.2% WB)

Anchor Residue Patterns

The model consistently identifies canonical anchors: - **p2 preferences**: L > M > I > V > Q > A > T (score: 1.0 to ~0.85) - **p9 preferences**: V > L > I > A > T > M (score: 1.0 to ~0.80) - **Non-canonical p2** (E, D, R, K, G, P, C) almost always NB - **C at p2** is the only non-canonical residue occasionally tolerated

Top Cross-Protein Candidates for Experimental Validation

Rank	Peptide	Protein	P(SB)	Clinical Relevance
1	LLGRNSFEV	p53 264-272	99.3%	Validated immunotherapy target
2	GILGFVFTL	M1 58-66	99.8%	Gold standard positive control
3	NLVPMVATV	pp65 495-503	99.9%	CMV vaccine target
4	RMFPNAPYL	WT1 126-134	100%	Leukemia vaccine target
5	IMDQVPFSV	gp100 209-217	100%	Melanoma vaccine target
6	SLLMWITQC	NY-ESO-1 157-165	99.2% WB	Cancer vaccine target
7	YMNGTMSQV	Tyrosinase 369-377	100%	Melanoma antigen
8	KIADYNYKL	Spike RBD 87-95	99.9%	COVID-19 vaccine candidate
9	ALMDKSLHV	MART-1 56-64	100%	Novel melanoma candidate
10	RMPEAAPPV	p53 65-73	100%	Mutation hotspot region

Methods

- **Model:** Deep FFN (360-180-90-45-3), MHCflurry 2.2.0 trained
- **Encoding:** BLOSUM62 (9-mer to 9x20 tensor)
- **Scan method:** Sliding window (stride=1) across full protein sequences
- **Thresholding:** SB if $P(\text{SB}) > P(\text{WB})$ and $P(\text{SB}) > P(\text{NB})$
- **Known epitopes:** IEDB-verified HLA-A*02:01 binders

Output Files

File	Contents
data/protein_epitope_scan_Ext1.h5	First 3 proteins (772 peptides)
data/protein_epitope_scan_Extended.h5	Extended 7,371 proteins (2,764 peptides)
models/FFN_Deep.h5	Trained model (91.9% accuracy)

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